BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, Lahore, 2021. Session 2017-2019 to 2020-2022. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The major cell infected by HIV is:

(A) B-lymphocytes
(B) Neutrophils
(C) Helper T-lymphocytes
(D) Basophils

Q2. Which is included in non-vascular plants:

(A) Homworts
(B) Whisk ferns
(C) Club mosses
(D) Horse tails

Q3. Incomplete or imperfect digestion is called:

(A) Ulcer
(B) Obesity
(C) Dyspepsia
(D) Botulism

Q4. In myocardial infarction, which organ is affected:

(A) Lungs
(B) Eye
(C) Kidney
(D) Heart

Q5. Zoogeography is study of distribution of what in nature:

(A) Animals
(B) Plants
(C) Trees
(D) Zoos

Q6. Which of the following is related to phytol:

(A) $C_{20}H_{39}$ (B) $C_{39}H_{20}$

(C) $C_{22}H_{40}$ (D) $C_{40}H_{22}$

Q7. An activated enzyme consisting of polypeptide chain and a cofactor is known as:

(A) Holoenzyme (B) Apoenzyme

(C) Alloenzyme (D) Co-enzyme

Q8. Shark liver oil is used in medicine as a source of vitamins:

(A) A and B (B) A and C

(C) A and D (D) A and E

Q9. It is a third mechanism to defend the body against the foreign invaders is:

(A) Skin (B) Mucous membranes

(C) Phagocytes (D) Immune system

Q10. Trypanosoma is transmitted by the bite of infected:

(A) Mosquito (B) Dragon fly

(C) House fly (D) Tsetse fly

Q11. Water makes how much percent of total cell weight in bacterial cell:

(A) 40% (B) 50%

(C) 60% (D) 70%

Q12. Nuclear fusion in basidium is followed by:

(A) Meiosis (B) Mitosis

(C) Budding (D) Binary fission

Q13. In bacteria when the division is in three planes it will produce which arrangement:

(A) Streptococcus (B) Tetrad

(C) Sarcina (D) Diplococcus

Q14. A respiratory pigment of blue in color called haemocyanin is present in animals of which phylum:

(A) Porifera (B) Coelentrata

(C) Mollusca (D) Arthropoda

Q15. One micrometer (µm) is equal to:

(A) 1×10^{-4} of a meter (B) 1×10^{-7} of a meter

(C) 1×10^{-6} of a meter (D) 1×10^{-9} of a meter

Q16. As a result of energy conversion during light dependant reaction, reducing and assimilatory power is formed in the form of:

(A) NADP
(B) ADP
(C) NAD
(D) $NADPH_2$ and ATP

Q17. Oxygen diffuses how many times more quickly in air than in water:

(A) 8 times
(B) 80 times
(C) 800 times
(D) 8000 times

Essay Type (Group-I)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: Session 2017-2019 to 2020-2022

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Write down the functions of proteins.

ii Define co-factor and activator.

iii What do you mean by lock and key method?

iv Differentiate between competitive and non-competitive inhibitors.

v Differentiate between septate and non-septate hyphae.

vi How fungi is economically helpful in food industry?

vii Differentiate between proterostomia and deuterostomia.

viii How locomotion takes place in annelids?

ix Define metamorphosis.

x How mammals have evolved from reptilian ancestors?

xi What is Rubisco? Write down its functions.

xii Write down the molecular formulae for chlorophyll "a" and "b".

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define biotechnology.

ii Define hydroponic culture technique.

iii What is cell fractionation technique?

iv Differentiate between microtubule and microfilament.

v What are amoebae? Give example.

vi What are kelps?

vii Give characteristics of red algae.

viii Define slime molds.

ix Define bryophytes.

x What is double fertilization?

xi Differentiate between granulocytes and agranulocytes.

xii What are blue babies?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Define binomial nomenclature. Give two examples.

ii Write down about the structure of plasmid in bacteria.

iii Write about three important ingredients of saliva.

iv Define symbiotic nutrition.

v How trapping and decomposition of insects occur in pitcher plant?

vi Write two properties of respiratory surfaces in animals.

vii Define larynx.

viii Differentiate between diaphragm and pleura.

ix What is asthma?

QSECTION - II. Note: Attempt any THREE questions.

Q5a. (Marks 4) What is biological method? Discuss it under following headings: (i) Theory (ii) Law

Q5b. (Marks 4) Write a note on types of immunity.

Q6a. (Marks 4) Describe secondary and tertiary structure of protein.

Q6b. (Marks 4) Write a note on ascomycetes. Also give importance of yeast.

Q7a. (Marks 4) Describe physical and chemical methods to control bacteria.

Q7b. (Marks 4) Describe land adaptations in bryophytes.

Q8a. (Marks 4) Write notes on smallpox and polio.

Q8b. (Marks 4) Draw and describe the Calvin Cycle.

Q9a. (Marks 4) What are plastids? Give their three types and explain one chloroplast in detail.

Q9b. (Marks 4) Discuss nutrition in insectivorous plants. (Any two)

GROUP – II

Objective Type (Group-II)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, Lahore, 2021. Session 2017-2019 to 2020-2022. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The reasoning that moves from general to specific called:

(A) Inductive
(B) Deductive
(C) Scientific
(D) None of these

Q2. The heterogenous group of compounds related fatty acids is called:

(A) Lipids
(B) Carbohydrates
(C) Proteins
(D) Water

Q3. Enzymes involved in photosynthesis are found in:

(A) Lysosomes
(B) Chloroplast
(C) Leucoplast
(D) Vacuoles

Q4. The process of taking in solid material by a membrane is called:

(A) Pinocytosis
(B) Exocytosis
(C) Phagocytosis
(D) Autophagy

Q5. Solanum tuberosum is the scientific name of:

(A) Onion
(B) Tomato
(C) Potato
(D) Garlic

Q6. Mesosomes are internal extensions of the:

(A) Cell membrane
(B) Cell wall
(C) Capsule
(D) Slime

Q7. Algae differ from plants in sex organs in algae are:

(A) Multicellular
(B) Acellular
(C) Unicellular
(D) None of these

Q8. Lovastatin is fungal product which lowers blood:

(A) Sugar
(B) Cholesterol
(C) Urea
(D) Calcium

Q9. All seed producing plants are called:

(A) Bryophytes
(B) Anthrophytes
(C) Pteridophytes
(D) Spermatophytes

Q10. Mammals have evolved from reptilian ancestors called:

(A) Cotylosaurs
(B) Echidna
(C) Opossum
(D) Archaeopteryx

Q11. Which one of the following is placental mammal:

(A) Echidna
(B) Kangaroo
(C) Bat
(D) Kingfisher

Q12. Chlorophyll "a" is:

(A) Yellow green
(B) Blue green
(C) Orange green
(D) Red green

Q13. Calvin cycle is also known as:

(A) C_3 pathway
(B) C_4 pathway
(C) C_5 pathway
(D) C_6 pathway

Q14. The first part of small intestine is called:

(A) Rectum
(B) Ileum
(C) Jejunum
(D) Duodenum

Q15. Respiratory pigment present in muscles is called:

(A) Myoglobin
(B) Haemoglobin
(C) Haemocyanin
(D) Globulin

Q16. The pathway involving system of adjacent cell walls throughout plant root is called:

(A) Symplast
(B) Apoplast
(C) Plasmodesmata
(D) Vacuolar

Q17. Single circuit heart is found in:

(A) Amphibians
(B) Reptiles
(C) Fishes
(D) Aves

Essay Type (Group-II)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i What is chemical definition of carbohydrates? Give its general formulae.

ii Define reversible inhibitors. Name its two types.

iii Write the induce-fit-model of enzyme action.

iv Write the function of penicillin and lovastatin.

v Name the fruiting body of Fungi, Ascomycota and Basidiomycota.

vi Describe co-factor and co-enzyme.

vii Define term protandrous and gemmule.

viii What is archaeopteryx? Give its two characters.

ix Name two super classes of vertebrates. Give example.

x Write any four characters of class osteichthyes (Bony fish).

xi What is Cytochrome? Give its role.

xii Define chemiosmosis.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i What is inductive reasoning, give one example?

ii Write briefly about hydroponic culture technique.

iii Why the plasma membrane is a differentially permeable membrane?

iv Differentiate between microtubules and microfilaments.

v Write two characters of Zooflagellates.

vi Write the functions of micronucleus and macronucleus in ciliates.

vii Write two characters of euglenoids.

viii How does conjugation occur in ciliates?

ix What is heterospory?

x Define double fertilization, in which plants it occur.

xi What is apoplast pathway?

xii Define imbibition in plants.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Define binomial system of nomenclature.

ii What are microaerophilic bacteria? Give one example.

iii What are leguminous plants?

iv Differentiate between intracellular and extra cellular digestion.

v What is antiperistalsis?

vi How aquatic plants obtain their oxygen?

vii What is a larynx?

viii What is diaphragm?

ix What is the main cause of lungs cancer?

QSECTION-II. Note: Attempt any THREE questions.

Q5a. (Marks 4) How biology has helped in increasing food production?

Q5b. (Marks 4) Explain various functions of blood in human.

Q6a. (Marks 4) Write short note on lipids.

Q6b. (Marks 4) Give detail of taxonomic status of fungi.

Q7a. (Marks 4) Describe characteristics of cyanobacteria.

Q7b. (Marks 4) Elaborate evolution of seed habit in plants.

Q8a. (Marks 4) Describe infection cycle of HIV.

Q8b. (Marks 4) Draw and explain glycolysis in detail.

Q9a. (Marks 4) Write a note on structure and function of plastids.

Q9b. (Marks 4) Write about food poisoning and obesity.